Windowed ANF Delivery: Packing the DB-Mixed Circuit into ANF Blocks

local_mixing project note

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Status: prototype-validated pipeline stage (python tools in `runs/ncdb_ab_20261001/`), production artifacts at $n = 64$ and $n = 128$ (`waist/final64/`, `waist/final128/`); Rust port planned per `docs/windowed_delivery_design.md` §7. All measurements cited here are recorded in that run's `RESULTS.md` and in `runs/exposure_report_20261003/REPORT.md`.

1 The current process

The delivery stage consumes the **stage-3 (DB-mixed) circuit** and produces the shipped artifact. It replaces the former stages 4–6 (split, crossing walk, fcompress/packing). Three steps.

1.1 Commuting reorder + window scheduling (`window_pack3.py`)

The gate order of the mixed circuit is re-scheduled within its exact commutation freedom, and simultaneously partitioned into contiguous **windows**. Scheduling and partitioning are one optimization because the reorder is where all the leverage is (§2.3).

Commutation model (MPMCT). Same-target writers commute; a reader pins against *all* later writers of the wire it reads. Pairwise RAW/WAR edges are unsound for this gate set; the scheduler therefore uses epoch conditions: per wire, a write counter; a reader of wire c is ready iff `writes_done[c]` equals its required epoch, a writer iff all earlier-epoch readers of its target have been emitted. Function preservation is by construction and is additionally verified (256 bit-sliced input samples, all wires) on the emitted order.

Value-aware planning. Before scheduling, the stage-2 reference circuit A and the mixed circuit B are simulated on 4,096 shared random inputs; every intermediate value is fingerprinted. This yields the **carried values** of B (values of A that re-appear on B's wires) and their **spans** (birth write $\rightarrow$ next write of the carrier wire). A value is *hidden* in the delivered artifact iff **none** of its spans contains a window boundary; hiding is therefore all-or-nothing per value, over all of its spans (typically $\sim$3, up to several hundred).

How the blocks are determined. The partition is *not* a greedy accumulation of gates; greedy filling is only the background activity. The window structure is driven by **interval reservations and atomic bursts**, in five steps:

1. **Interval extraction (per span).** For each span (birth write $b \rightarrow$ killing write e) of each carried value, compute the span's *two-sided tainted closure*: a forward taint sweep from b (a

gate joins if it reads a tainted-written wire or writes a tainted-read wire — the sound any-tainted rule, since pairwise RAW/WAR edges are unsound for MPMCT), intersected with a backward taint sweep from e. The result is the minimal gate set that must share a window for that span to be hidden — its *members*. A span whose member set exceeds `W_cap` gates disqualifies (one rescue is attempted: extend to the next-next writer of the carrier wire). A value is **winnable** iff *all* of its spans have member sets $\leq$ `W_cap`; values with any raw span longer than `span_cap` are disqualified outright.

2. **Readiness bookkeeping (epochs).** Every gate carries an unmet count: a reader of wire c becomes ready when c reaches the write-epoch the original order gave it; a writer of wire t when all earlier readers of t are emitted. This is the exact commutation freedom — the scheduler can emit any ready gate at any time without changing the function.

3. **Drain phase (the greedy part).** Intervals are processed by ascending end position. Between bursts the scheduler emits **filler**: ready gates that belong to *no* living interval (protection count zero). Filler fills windows with material whose placement is exposure-irrelevant.

4. **Atomic bursts + value-level accounting.** When an interval's turn comes, the window is first checked: if the remaining gate capacity is below the interval's size (+ a pad of 4) or the composed ANF is above $\frac{3}{4}$ of the monomial budget, the window is closed *pre-emptively* so the burst is never split by a scheduled close. Then **all member gates of the interval are scheduled consecutively** (dependencies pulled from the ready heap as needed) inside the one window. Hiding is all-or-nothing per value: the first touched span commits the value; if any span dies — a window closed mid-burst, or a member was consumed earlier as filler/another burst (`resumed_later`) — **all sibling intervals of that value are killed** (`kill_value`) and their reserved gates are released to the filler pool immediately, so no capacity is spent on half-hidden values. After each burst, **contagion** completes any committed interval with ≤ 24 remaining members in place while the window is still open.

5. **Budget-adaptive close.** The close condition measures the *real* deliverable cost: the window's sparse ANF is composed incrementally gate-by-gate during scheduling, and the window closes when the monomial count exceeds the soft budget `B_mono` (hard $2 \cdot$ `B_mono` inside a burst) or the gate count exceeds the soft target G (hard $2G$ inside a burst). Every interval death is attributed (`drain_close` / `burst_close` / `resumed_later` / `sibling`), which is how scheduler losses are diagnosed (§2.4).

After the main pass, remaining gates are emitted by readiness, the final window closes, the emitted order is re-verified functionally, and the ANF emission stage (§1.2) re-composes each window independently against the larger emission budget with per-window verification.

1.2 ANF emission (`anf_compose.py`)

Each window is delivered as the canonical algebraic normal form of its touched wires: for every wire written inside the window, the exact GF(2) polynomial of its value at the window's end as a function of the window's *input* wire values. Composition is sparse and exact, with true GF(2) cancellation (per-monomial toggle — symmetric-difference dedup is a measured bug, caught by re-verification). Monomials and wires are canonically ordered; the within-window gate text, gate count, gate order, and all intermediate spellings are discarded.

Budget: 16,384 monomials per window. Measured usage is far below it: median 184 (n=64) / 207 (n=128) monomials per window, p90 $\approx$ 290–430, maximum 7,673, **zero overflows** at both

scales. Every window is functionally re-verified against the gate-level window on the shared sample set before serialization.

1.3 Serialization

The artifact (`deliverable.bin`) is the sequence of window ANFs in a binary format (wire ids, monomial lists over window inputs). Nothing else ships: no gate list, no boundary metadata beyond the windows themselves, no stage provenance.

	$n = 64$ ($\|C\| = 1{,}000$)	$n = 128$ ($\|C\| = 3{,}000$)
stage-2 circuit	771,029 g / 256 w	2,240,757 g / 512 w
stage-3 (mixed) input	1,648,679 g	4,803,546 g
windows	10,702	28,986
ANF payload	24.6 MB (×3.88 vs gate list)	93.7 MB (×5.06)
`deliverable.bin`	27.5 MB	103.1 MB
xz-compressed	4.52 MB	15.3 MB

Evaluation cost (python reference evaluator, 1,024-lane bit-sliced): 0.68 / 1.70 ms per input in gate form, 0.85 / 2.54 ms in ANF form, bit-exact agreement between the two forms at both scales. Compressibility is 6.1–6.7× under xz — and the compressed ANF is *smaller* than the compressed gate list, i.e. the ×4–5 raw inflation is redundancy of the format, not information added.

1.4 Full production parameters, both scales

Stage-2 knobs are identical at both scales (the managed embedded-masking preset): K=2 band wires per LGI, `max_open` 3 / `min_open` 2, quad-fire, balanced masks and balanced seed, rich band seed (`seed_fill_nl` 4), feistel slice guards (all gates $\leq$ 2 controls), carrier shuffling dose 8 with **persistent migration** (`EMBEDDED_MASKING_SHUFFLING=8`, `..._RETURN_HOME=0`). Stage-3 flags are identical: `--tdp --db-mixing --profile 2,2,2,2,2 --p-db 1.0 --p-comp 1.0 --p-any 0.1 --db-max-degree 9 --db-max-span 30 --db-wire-terms 1024 --db-total-terms 2048`, regular frozen store `frozen_m1_m11` (`--p-comp`/`--p-any` are retired since the four-stage cleanup: they were inert at `--p-db 1.0`, and current builds reject them), moves = $\lceil 2 \cdot 2 \cdot 1.3 \cdot (\text{stage-2 gates}) \rceil$. Packing and emission are identical: `W_cap` 128, G 128 (hard 256), `B_mono` 6,144 (hard 12,288 in a burst; $\frac{3}{4}$ pre-burst guard), `span_cap` 3,000, contagion threshold 24, pad 4; emission budget 16,384 monomials/window; 4,096 fingerprint lanes. Fresh CSPRNG seeds per run, recorded in each run directory.

parameter	$n = 64$ run	$n = 128$ run
source C	64 wires, 1,000 gates	128 wires, 3,000 gates
sandwich ($\|D\|$, s, slice gates)	1,000 / 384 / 1,280	3,000 / 896 / 2,560
sandwich circuit	2,832 g / 128 w	7,920 g / 256 w
stage-2 output	771,029 g / 256 w	2,240,757 g / 512 w
mixer moves	4,009,351	11,651,936
stage-3 output (wall time)	1,648,679 g (∼26 min)	4,803,546 g (∼76 min)
carried / winnable	145,481 / 131,226 (90.4%)	426,891 / 382,071 (89.5%)
hidden by construction	125,592 (95.7% of winnable)	337,934 (88.4%)
windows	10,702	28,986
window monomials med/p90/max	194 / 301 / 7,682	207 / 332 / 2,846
exposed distinct at boundaries	23,347 (16.0% of carried)	89,072 (20.8%)

2 Rationale, tradeoffs, and what the process optimizes

2.1 The threat model that motivates windowing

Every exposure metric in the 2026-10 program — per-value visibility, the t^* exposure census (exact affine predictors), the equality-cluster attack — quantifies over **cuts of the delivered spelling**: the adversary reads wire values at positions between delivered blocks. Two measured laws made gate-granular delivery look fundamentally leaky:

- **Reader law**: any value with ≥ 1 bare reader is ∼100% visible under every-gate cuts (measured 3.2 / 55.4 / 63.8 / 100.0% visibility at fanout 0/1/2/≥3 on the TDP, stage 3).
- **Closure law**: function preservation forces interior basis changes to close (the second bracket undoes the first), so **no self-closing interior move can raise the min-over-cuts predictor weight at any dose** — a CNOT/transvection twist sweep that bloated the circuit ×37 left t^*=1 mass at 84%. The interior move system contains no weight-raising operator; the t^* spectrum belongs to the circuit ends.

Windowing attacks the quantifier instead of the values: **a window has no interior cuts.** A value is exposed only if it is alive across a window boundary. Deleting cuts (a) makes read values hideable for the first time — the reader law presumed a cut between every pair of gates; (b) sidesteps the closure law — nothing is rewritten between cuts, the cuts are deleted; and (c) annihilates all within-window text: templates, k-grams, gate provenance, stage-2 vocabulary signatures, and the gate count itself vanish into a canonical polynomial. What remains observable is exactly: values alive at boundaries, the boundary equality-linkage graph, and the outputs.

2.2 The objective function

The packer maximizes the number of **distinct carried stage-2 values hidden** — ground truth measured by fingerprint simulation on the emitted order, not a proxy. Secondary structure it respects:

- value-level all-or-nothing (a value with one exposed span is exposed; partial hiding is worthless);
- the C-view invariant: the source computation's interior must stay at **zero** exposure (measured: the output-only floor — exactly the 64 / 128 outputs visible, all t^*=1 — holds through the entire delivery stage at both scales);

- ANF budget and window caps (digestibility and size).

2.3 Where the leverage is: reorder, not boundary placement

Measured on the TDP: delivered-order carrier spans are long and waists are flat — optimal boundary placement in the *given* order is approximately uniform placement, worth almost nothing. But a value's minimal bare residency under commuting reorder is its **DAG interval**, median 6 gates on stage-3 material — ×21 **shorter** than its spelled span. The entire mechanism is: reorder to compress each value's bare residency to its DAG interval, then drop windows around the intervals.

2.4 The window-size tradeoff

The one real tradeoff surface is window size G:

- **Hiding rises** $\sim$+14 **points per doubling of** G (measured sequence, migrated material: 46% hidden @G=64, 58% @G=128, 72% @G=256 — 5.2% of the full stage-2 vocabulary still exposed at G=256).
- The **ANF budget never binds**: median monomial usage is $\sim$200–400 of multi-thousand budgets even at 512-gate windows, zero overflows anywhere in the program. The limit on G is architectural, not combinatorial: boundary count vs the circuit ends, the re-synthesis-hardness floor (very small windows are canonically re-synthesizable; very large windows reduce the boundary count toward the ends), and delivery size ($\approx$ ×12 at G=512).
- Scheduler quality, not material, is the current binding constraint: on migration-refreshed material, 93% of carried values are winnable while v3 hides 72–73% of winnable; the gap is attributed by the death counters to burst-time closes and sibling cascades (scheduler v4, value-unit planning with conflict-graph pre-grouping, is the specified next step; estimated ceiling 94%).

Packer evolution, same ground truth (% of carried values hidden, stage-3 input, W=20 era → G-era): naive uniform windows 5% → v1 interval-burst scheduler 21% → v2 value-level all-or-nothing 32% → v2.1 35% → v3 budget-adaptive 46/58/72% at $G = 64/128/256$.

2.5 Costs accepted

- **Size** ×4–5 raw (×$\approx$0.9 after compression, §1.3): the format buys erasure, not compression.
- **Boundaries are new visible structure**: the window lattice itself is observable. Measured consequence: a value recurring ≥ 5 times among the carried values is *always* boundary-exposed (deterministic at both scales; non-exposed maximum multiplicity is 4). This "recurrence law" shifted the program's focus upstream to stage-2 carrier hygiene (the celebrity program: migration, hot-targeting, threshold retirement), since no boundary placement can hide a value whose spans tile the stream (celebrity liveness was measured at $\sim$110 of 256 wires at *every* cut position).
- **Affine-metric caveat**: all exposure numbers quantify exact affine predictors at boundaries; the approximate/nonlinear-predictor flank is an open thread of the exposure program (known empty for the QC-style detector: agreement ≥ 0.98 coincides with exact equality).

3 Why stages 4–6 were abandoned

The decision rests on the stage-wise exposure study (nine arms × four stages, random family and TDP, 2026-10-03) plus the direct input-choice measurement of the windowing program (2026-10-04).

3.1 What stages 4–6 actually bought (measured)

Per-stage visibility % [$t^*\geq 2$ share %], random-family arms, every-cut grids (excerpt; full table in `runs/exposure_report_20261003/REPORT.md`):

arm	s3 mix	s4 split	s5 crossing	s6 fcompress
random f=2	85.1 [1.5]	84.0 [8.3]	76.1 [8.9]	79.2 [5.8]
f=10	79.5 [1.7]	79.0 [5.1]	76.0 [6.1]	76.3 [4.3]
eq. arity-2	82.6 [1.3]	81.6 [4.5]	78.7 [6.0]	79.4 [4.6]
twists p=.05	87.0 [2.2]	87.0 [5.8]	82.7 [7.8]	83.9 [5.8]

Stage signatures, replicated at TDP scale:

- **Stage 4 (split)** is a visibility-neutral t-raiser: it moves 4–8 points of t^*=1 mass into small parities and erases nothing. In carrier terms it *adds* exposure surface: 105.7k → 107.9k carried values on the TDP.
- **Stage 5 (crossing)** was the only post-processing eraser: $-3\ldots-5$ points of visibility in every arm (105.7k → 89.8k carried on the TDP). This is the entire measured security content of the old post-3 pipeline.
- **Stage 6 (fcompress)** partially *refunds* stages 4–5: $+1\ldots+3$ points, parities collapse back. Its packing component is exposure-neutral (pack-only ≡ crossing to ±0.1; pack-then-compress ≡ compress-then-pack, order flip null at +2% size).
- At TDP scale the whole chain is flat: stage-2-view visibility 26.5 → 26.5 → 25.6 → 27.2/27.3% across s3 → s6. The C-view output-only floor is owned by stage 2 and never moves.

So the old pipeline's net effect was: one eraser (crossing), one neutral t-raiser (split) whose gains the compressor refunded, at the price of large inflation and run time.

3.2 The direct comparison: windowing s3 vs windowing s5

The windowing program's first decision was its input. Measured:

- **Crossing welds the dependency structure.** Its ×3.45 inflation stretches the median DAG interval from 6 gates (s3) to 26 (s5) — ×4 — so the reorder lever that windowing lives on is largely destroyed. Per-value hideable ceilings at matched window size W=20: 69.6% of carried values (s3) vs 46.9% (s5).
- **Crossing's erasure does not pay for the welding.** In absolute terms at W=20: ∼32k exposed values on the s3 ceiling vs ∼48k on the s5 ceiling — windowing the raw mixed circuit strictly dominates windowing the crossed one, despite crossing's 15% carrier erasure.
- **Split only adds carriers** (§3.1), and the walk family generally is exposure-hostile: the order-preserving greedy walk measured 97.4% exposure against its own 79.4% floor — verbatim-dressing passes are exposure-*maximal*.

Against these, a single windowing pass hides 58–72% of carried values (G=128–256) — an order of magnitude beyond crossing's $-3 \ldots -5$ points — while *deleting* within-window text rather than respelling it, at lower wall-clock cost than the crossing walk it replaces.

3.3 Division of labor after the replacement

- **Stage 2** — semantic hiding: which values ever exist bare (data path at zero bare reads; band carrier hygiene: lifetime, migration, dedup/retirement, seed density).
- **Stage 3** — un-invertible respelling, dilution, DB-form digestion.
- **Windowed ANF delivery (new stage 4)** — cut deletion: *where* anything is observable at all, plus canonical within-window respelling.
- **Circuit ends (sandwich/slicing)** — the t^*-spectrum of the survivors (interior moves provably cannot move it).

4 Production reference results (pipeline v2.1, final64/final128)

End-to-end: classic sandwich → embedded masking (rich band seed, persistent carrier migration dose 8, read-pressure partners, feistel slice guards) → db_mixing → window-pack G=128 → per-window sparse ANF. Function checks green at every stage, both scales; all windows re-verified; zero budget overflows; fresh CSPRNG seeds recorded in the run directories.

leakage parameter	$n = 64$	$n = 128$
C-view (the secret)	6.4% = exactly the 64 outputs, t^*=1	4.3% = the 128 outputs (+2 interior t^*=4)
stage-2-view visible (exact lattice)	27.3% (t^*=1: 22.1, $t^*\geq$2: 5.2)	28.4% (23.6 / 4.7)
distinct values exposed at boundaries	23,347 = 16.0% of carried = 3.8% of vocab.	89,072 = 20.8% / 5.0%
class uniformity (data/mid/band)	26.5 / 27.3 / 27.7%	28.3 / 27.2 / 28.7%
non-exposed max multiplicity	4	4
max boundary equality cluster (proxy)	342	397

The residual exposed mass is dominated by high-recurrence band vocabulary (the celebrity class, 330/966 values of multiplicity $\geq$ 257), which is a **stage-2 birth phenomenon**, not a packing phenomenon: boundary redrawing was measured infeasible (celebrity spans median 494 / p95 2,505 packed gates, liveness ∼110 wires at every cut position), and the fix in progress (threshold retirement of band carriers at the emitter) operates upstream of this stage. The delivery stage's own contract — C-view at the output-only floor, within-window annihilation, recurrence-law behavior at boundaries — is identical with and without that fix.

5 Artifacts and tools

- Design doc: `docs/windowed_delivery_design.md` (architecture + the stage-2 band-subsystem spec).

- Tools: `runs/ncdb_ab_20261001/window_pack3.py`, `anf_compose.py`, `boundary_cluster.py`, `waist/recurrence2.py`, `waist/anf_dump_eval.py`, `wire_balance.py`; exposure scans via `hmap_affine --exposure --g-cuts` (boundary-lattice mode).
- Production artifacts: `runs/ncdb_ab_20261001/waist/final64/` and `final128/` (stage-2/3 circuits, packed order + bounds, deliverable.bin + compressed forms, full census logs, seeds).
- Measurement record: `runs/ncdb_ab_20261001/RESULTS.md`; stage-wise study: `runs/exposure_report_20261003/REPORT.md`.